



**UNIVERSITY OF
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PUSL3190 Computing Project

Abstract

SPAREHUBLK

Supervisor: Pavithra Subhashini

Name: Weerasundarage Weerasinghe

Plymouth Index Number: 10954915

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This project focuses on the problems faced by vehicle owners and mechanics when searching for spare parts using general online marketplaces. Most existing platforms are not designed specifically for vehicle parts, which makes it difficult to find the correct item, leads to mismatched purchases, and wastes time. To address this issue, this project aims to develop an AI based web platform for buying and selling vehicle spare parts in Sri Lanka.

The system was developed as a web application using the MERN stack, including MongoDB, Express, React, and Node.js. The platform allows users to register, log in, list spare parts, and search using keywords. Additional features such as image based search, a chassis number decoder, price suggestion, and a chat system are included as intelligent components. The development was carried out step by step, starting with core features and gradually integrating advanced functions. The system was tested using different user scenarios to evaluate functionality and usability.

The results show that the system can support users in listing and searching spare parts more efficiently compared to existing platforms. It reduces the difficulty of identifying correct parts and improves the searching process. Features such as structured listings, chat communication, and premium seller shops improve user interaction and trust. Some advanced features are still being improved, but the main system functions are working successfully.

In conclusion, the proposed system provides a more focused and user friendly solution for spare parts trading. It helps users save time, reduce mismatches, and improve the overall buying and selling experience compared to current online marketplaces.